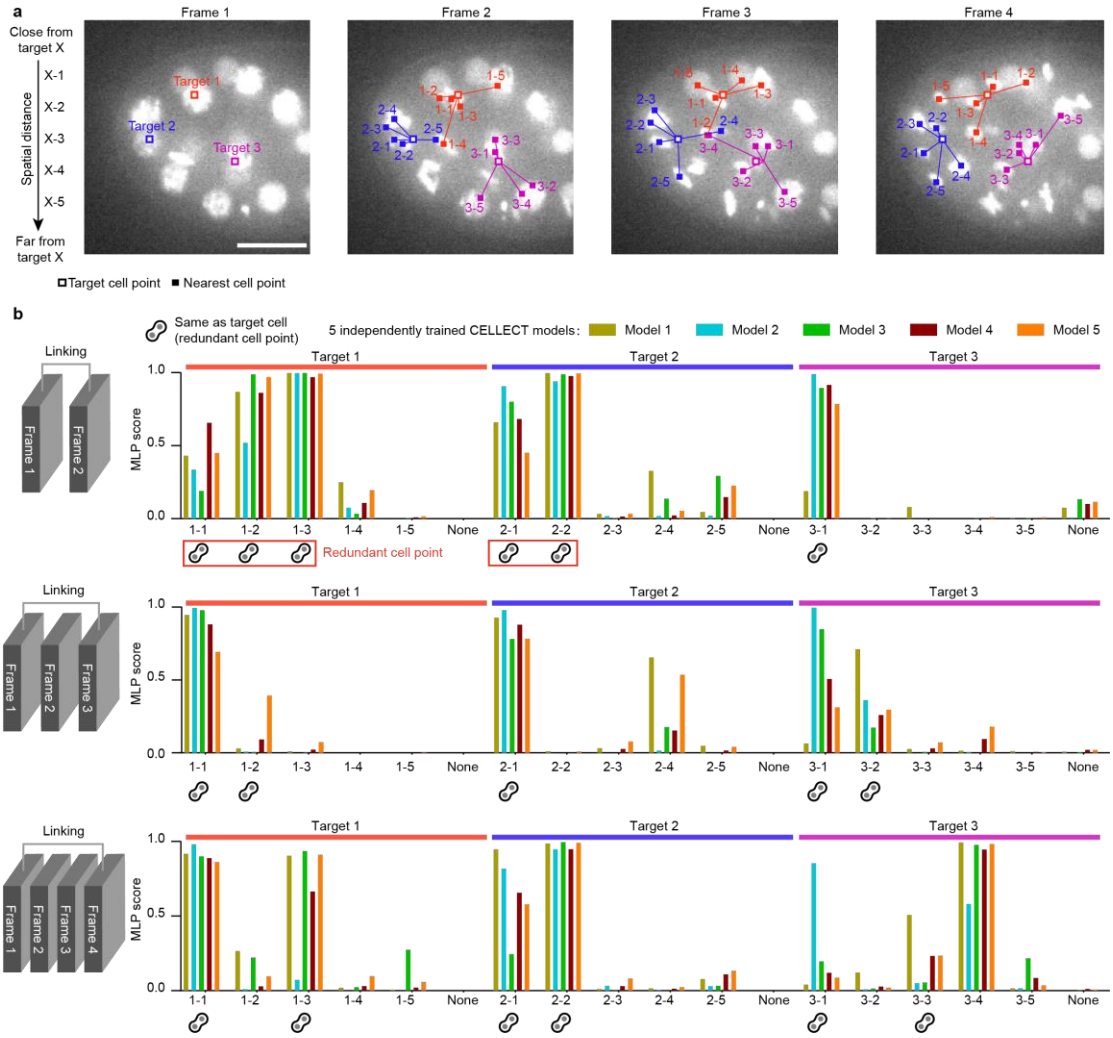



**Supplementary Figure 14 | Evaluation of the reproducibility of cell linking during different imaging frame rate.** To evaluate the reproducibility of CELLECT's cell linking under different imaging frame rate, we selected 3 representative target cells (Target #1–3) from mskcc-confocal dataset #2 and selected 5 nearest spatial neighbors at frames t+1, t+2, and t+3 for testing. **a**, Location of the target cells at time t (frame 1) and the locations of candidate cells across the 3 subsequent frames. **b**, Similarity scores predicted by 5 independently trained CELLECT models for the 5 candidates. To ensure consistent input across models, redundant point removal was not applied intentionally, resulting in occasional duplicate redundant cell points (e.g., Target #1 and #2 at frame 1 linked to frame 2). This configuration increases linking difficulty by expanding the temporal gap, thereby simulating more complex multi-frame association scenarios. While correct links often correspond to the nearest candidate in adjacent-frame conditions, longer intervals such as frame 1 to frame 3 or frame 4 require the model to rely more on embedding features for accurate decisions. The results demonstrate that CELLECT produces consistent linking performance across different frame intervals. However, under low imaging frame rate and in the presence of cell divisions (e.g., frame 1 linked to frame 4), a slight drop in accuracy is observed. All 5 models were independently trained on mskcc-confocal dataset #3. Scale bar: 10 μm.